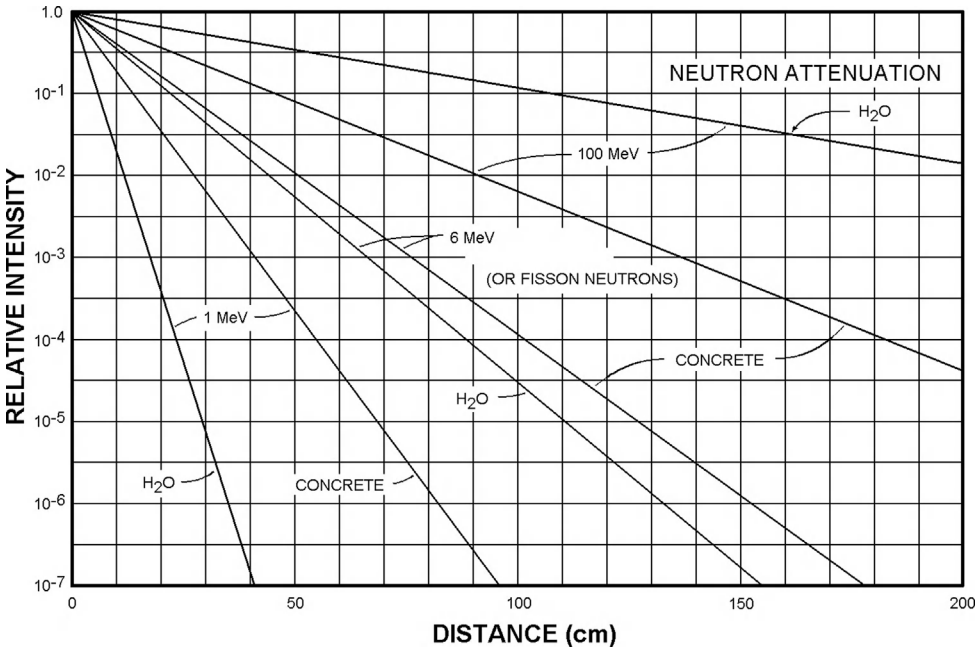


Figure 7.21 Shielding thickness necessary to reduce a neutron beam in water and concrete. The 6 MeV lines can be used for fission neutrons. *(From Nuclear Data Tables.)*

Equation (7.27) is not directly applicable for neutrons. For an estimate of required shielding we can use diagrams like that in Figure 7.21, which shows the attenuation of neutrons of three different energies in concrete and water, the most common neutron-shielding materials. It is necessary also to take into account the γ-rays emitted in neutron capture, which increases the shielding thickness required. (Please note that the order of absorption change as the energy of the neutron is increased. This is due to the energy dependence of the dominating cross sections, especially for the heavier atoms present in concrete.)

7.8. ANALYTICAL APPLICATIONS OF RADIATION ABSORPTION

In previous sections of this chapter we have shown how different particles emitted in nuclear reactions are stopped in matter without causing nuclear reactions (nuclear reactions are treated in Ch. 10 and 11). If the property of the particle is well defined (e.g. mass, charge, energy, etc) its interaction with the absorbing material can be well predicted, provided the composition of the absorbing material also is well defined. Conversely, the composition of the absorbing material (a "sample") can be determined from studying the absorption process (the "irradiation"). For example, atoms of the absorber (sample) may be knocked out and can be collected for analyses. When electrons

are knocked out from the atomic shells of the absorber (sample) atoms, either the energy of these electrons may be analyzed, or the energy of the electromagnetic radiation emitted when the shells are refilled. These energies are characteristic of the sample and can thus be used for its identification. There are many analytical applications based on these principles, the most important ones will described in this section.

7.8.1. SIMS (secondary ion mass spectrometry)

When heavy ions of energy largely exceeding the chemical binding energies, but with energies much lower than needed to cause nuclear reactions, hit a surface then atoms of this surface are *sputtered* out. These atoms, or actually ions, can be introduced in a mass spectrometer to determine the exact masses and mass/charge ratios, from which the element is identified. This is the bases for the SIMS analytical method for studying surfaces, particulary semiconductor surfaces. By bombarding with O^{2+} or Cs^{+} of ≤ 10 keV most surface elements can be detected (in fortunate cases down to the ppb range) with a resolution of the order 100−200 Å.

7.8.2. PIXE (proton or particle induced X-ray emission)

In the PIXE technique high energy protons (or heavier ions) are used to irradiate a thin sample (say 0.1 mg/cm^2). The probability for expulsion of an electron followed by the emission of a K-X-ray decreases with increasing Z of the sample; e.g. for 5 MeV protons the reaction cross section is ca. 10^{-25} m^2 for $Z = 10$ to 20 and about 10^{-28} m^2 (1 *barn*) for $Z = 50$; cf. Table 7.1. It can be shown that the reaction probability has a maximum when the incident particle has a velocity equal to the Bohr orbital velocity of the electron. Though the sensitivity decreases with Z elements up to Pb can be determined. The technique has primarily been used to determine trace elements in environmental and biological samples, see Figure 7.22. It has also been used by several Mars landers to analyse the composition of various rocks and minerals on the surface of Mars.

7.8.3. ESCA (Electron spectrometry for chemical analysis)

High resolution β-spectroscopy can be used to determine chemical properties. A sample is irradiated with mono-energetic photons of $E_{\mathbf{h}\nu}$, leading to the emission from the sample surface of photoelectrons. The relevant equation is

$$E_{\mathbf{h}\nu} = E_{\mathrm{be}}(\mathrm{X,Y}) + E_{\mathrm{e}} \tag{7.29}$$

where E_{e} is the kinetic energy of the emitted electrons, which can be determined very accurately (presently to about 0.01 eV) in the magnetic spectrograph; *photoelectron spectroscopy*. This sensitivity is much greater than chemical binding energies, E_{be}(X,Y), where X refers to an atom in compound Y. The probability for ejection of photoelectrons increases with decreasing photon energy (Fig. 7.16 and 7.19) and therefore low energy X-rays are used as a source.

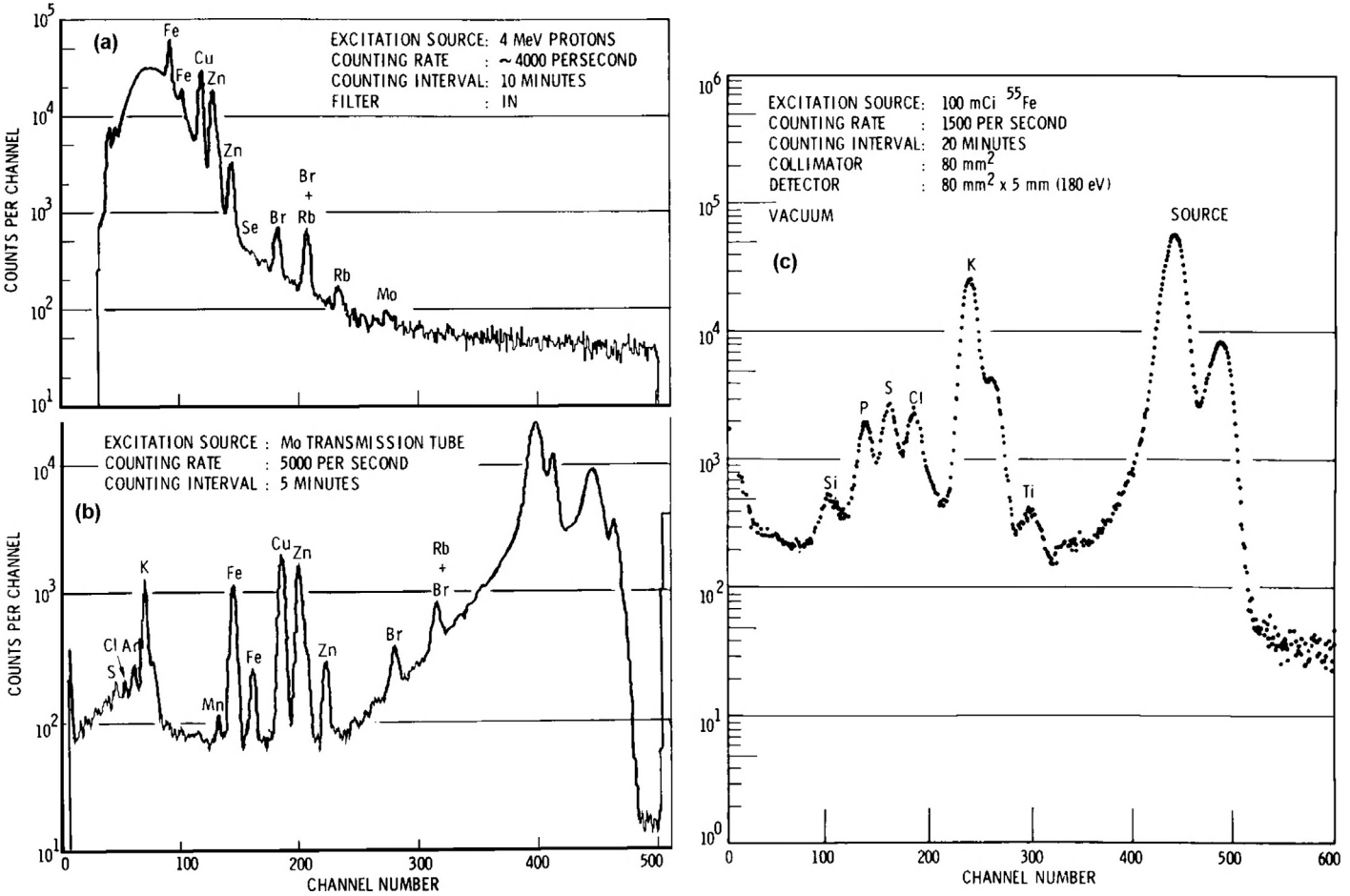


Figure 7.22 Fluorescent spectra obtained on bovine liver with three different excitation sources: (a) 4 MeV protons (PIXE), (b) (conventional) molybdenum transmission tube, and (c) radioactive ^{55}Fe.

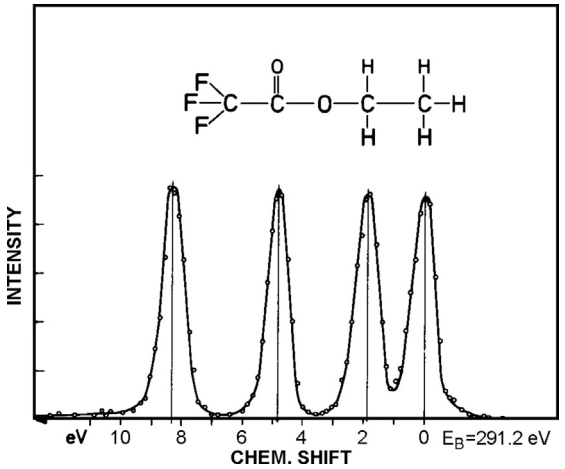


Figure 7.23 ESCA spectrum of trifluoroacetate.

Although it is the outermost (or most weakly bound) electrons which form the valence orbitals of a compound, this does not leave the inner orbitals unaffected. An outer electron (which we may refer to as e_L) of an atom X_1 which takes part in bond formation with another atom X_2 decreases its potential, which makes the inner electrons (which we may call e_K) more strongly bound to X_1. Thus $E_{be}(e_K)$ increases by an amount depending on $E_{be}(e_L)$. Although this is a somewhat simplified picture, it leads to the practical consequence that the binding energy of e_K, which may be in the 100−1000 eV range, depends on the chemical bond even if its orbital is not involved directly in the bond formation. Figure 7.23 shows the ESCA spectrum of trifluoroacetate. Since the largest chemical shift, relative to elementary carbon, is obtained for the most electronegative atoms, the peaks refer to the carbon atoms in the same order as shown in the structure.

7.8.4. XFS (X-ray fluorescence analysis)

If a sample containing atoms of a particular element (e.g. Ag) is irradiated with photons of energy high enough to excite an inner electron orbital (e.g. the K_α orbital in Ag at 22.1 keV), X-rays are emitted in the de-excitation. If the photon source is an X-ray tube with a target made of some element (Ag in our example), the probability is very high that the K_α X-ray emitted from the source would be absorbed by the sample atoms and re-emitted (*fluorescence*). (This is an "electron shell resonance absorption" corresponding to the Mössbauer effect, §7.8.5, although the width of the X-ray line is so large that recoil effects can be neglected.) The spectrum of the scattered X-radiation (or, more correctly, photon radiation emitted from the sample) is referred to as the *X-ray fluorescence spectrum*. It contains lower energy radiation including K_α radiation

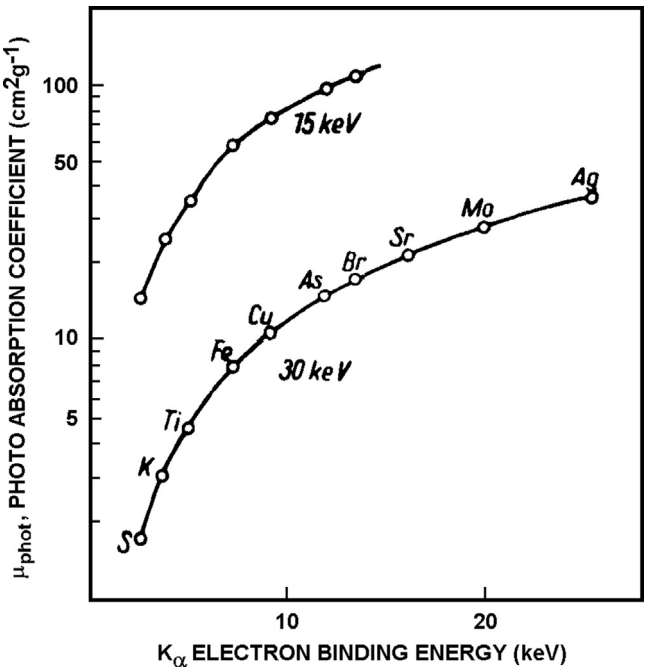


Figure 7.24 Photoelectron absorption coefficients at K edges for 10 and 30 keV γ-rays as function of absorber material. The photo effect is the dominating absorption mode.

emitted by atoms of lower atomic number. The height of these other peaks is lower because of a lower reaction cross-section (Fig. 7.24). Figure 7.22 illustrates an XFS analysis of a biological sample.

X-ray fluorescence analysis using vacuum tube sources have become a well-established analytical technique in the last decade. Nuclear interest stems partly from the possibility of using radioactive sources for stimulating X-ray fluorescence in a sample. These sources can be classified depending on the mode of production of the X-rays:

(i) γ-ray sources: decay between closely located nuclear energy levels, e.g. a 59.5 keV γ emitted in the α-decay of ^{241}Am: also "broad spectrum" γ-sources like ^{125}I are used.

(ii) X-ray sources: (a) radiation emitted in rearrangement of electron orbitals following α- or β-decay (primary X-rays), e.g. 11.6—21.7 keV uranium L-X-rays from ^{238}U formed in the α-decay of ^{242}Pu, or 41.3—47.3 keV europium K-X-rays from β-decay of ^{153}Gd; (b) irradiation of a target with α-, β-, or γ-radiation leading to ionization and excitation of the target atoms and its de-excitation by X-ray emission.

(iii) Bremsstrahlung sources, e.g. T in titanium, or ^{147}Pm in aluminum (Fig. 7.10).

An important advantage of radioactive X-ray fluorescence sources is that very small instruments requiring no (X-ray) high voltage or current can be designed for field applications. Such instruments are used in geological investigations (bore holes, mineral samples, etc.), in-line mineral analysis (Zn, Cu, and other ores in flotation cycles, Ca in cement raw-material, etc.), on-line analysis of surface depositions (Zn on iron sheets, Ag in photographic emulsions, etc.). Figure 7.22 shows fluorescent spectra obtained on bovine liver with three different excitation sources: (a) 4 MeV protons (PIXE), (b) (conventional) molybdenum transmission tube, and (c) radioactive ^{55}Fe.

7.8.5. Mössbauer effect

According to the wave model of the atom, electrons in the innermost orbitals have a finite probability of existence within the nucleus. These electrons interact with the nuclear charge distribution, and thereby affect the nuclear energy levels (cf. §11.3.3). The extent of the effect depends on the exact properties of the electron orbitals involved, which vary with different chemical compounds. Therefore a γ-ray emitted from an isomeric state of an atom bound in one chemical compound may have a slightly different energy than from the same atom bound in another compound. This difference, referred to as the *isomer* (*energy*) *shift*, is extremely small, only about 10^{-10} of the energy of the emitted γ. Nevertheless, it can be measured by a technique developed by R. Mössbauer. The fundamental physics involved and technique used is well illustrated by Mössbauer's original experiment. Mössbauer placed an ^{191}Os source about a half-meter from a γ-ray detector A as shown in Figure 7.25. An iridium foil absorber was placed between the source and detector so that some of the photons of 129 keV energy from the ^{191}Os were absorbed by the iridium atoms in the foil, exciting these atoms from the ground state

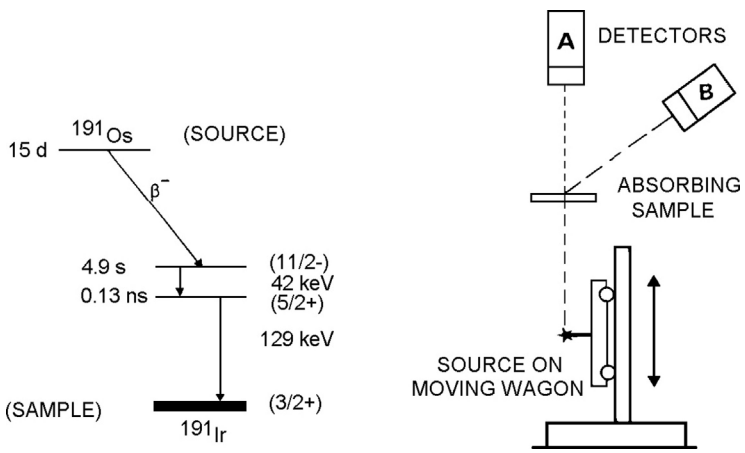


Figure 7.25 Decay scheme of ^{191}Os and principle of a Mössbauer experiment.

(3/2+) to the 5/2+ state. Because of the short half-life of the latter state it immediately decayed, re-emitting the γ-ray. The emission was isotropic, i.e., occurs in all directions. The result was a reduction in intensity measured by detector A but an increase in the count rate in detector B.

The conditions for such a *nuclear resonance absorption* are very stringent. Using the Heisenberg relationship, Eqn (5.89), it is possible to estimate the half-value width of the 129 keV peak to be 5×10^{-6} eV. Eqn (5.48) can also be used to calculate the iridium atom recoil energy. It is then found to be 46×10^{-3} eV. Thus the γ-ray leaves the source with an energy of $(129 \times 10^{3} - 46 \times 10^{-3}$ eV). Also, in order for the 129 keV γ-ray to be absorbed in ^{191}Ir, it must arrive with an excess energy of 46×10^{-3} eV to provide for the conservation of momentum of the absorbing atom. Thus there is a deficit of $2 \times 46 \times 10^{-3}$ eV, which is very different from the value of the very narrow energy width of the γ-ray. Consequently, no resonance absorption can take place.

The limitation posed by the recoil phenomenon can be circumvented. If the source and absorber atoms are fixed in a crystal, the recoil energy may be insufficient to cause bond breakage. The energy is absorbed as an atomic vibration in the crystal, provided the quantization of the vibrational states agree exactly with the recoil energy. If not, which is often the case, the absorber atom stays rigid in the lattice, and the recoil energy is taken up by the whole crystal. In this case it is necessary to use the mass of the crystal in Eqn (4.34) rather than the mass of a single atom. Under these circumstances the recoil energy becomes infinitesimally small for the emitting as well as the absorbing atom; this is called *recoilless absorption*. The probability for recoilless absorption is improved if the source and absorber are cooled to low temperatures.

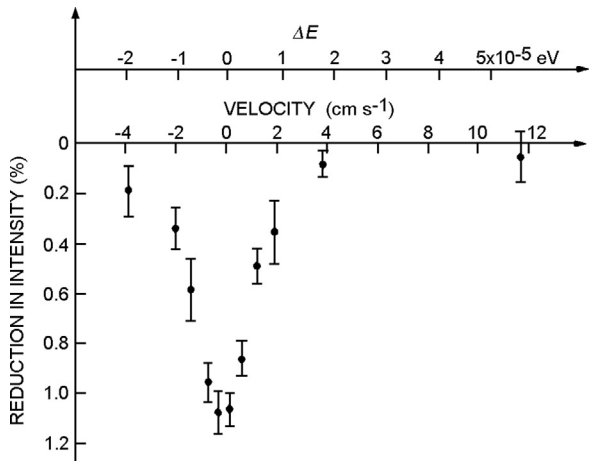


Figure 7.26 Mössbauer spectrum of ^{191}Ir metal. *(From R. Mössbauer.)*

The data of Figure 7.26 were obtained by recoilless absorption in osmium metal containing ^{191}Os (source) and Ir metal, both cooled in cryostats. By slowly moving the source (with velocity v) towards or away from the absorber (see Fig. 7.25), some kinetic energy ΔE_γ is added or subtracted from the source energy E_γ as "detected" by the absorber (Doppler effect). The energy and velocity relationship is given by the Doppler equation

$$\Delta E_\gamma / E_\gamma = v/\mathbf{c} \tag{7.30}$$

The velocity is shown in the Figure, where a value of v of 1 cm s^{-1} corresponds to 4.3×10^{-6} eV. The half-value of the γ-peak is found to be about 20×10^{-6} eV, i.e. a factor 4 times higher than calculated by the Heisenberg relationship. This is due to Doppler broadening of the peak as a consequence of some small atomic vibrations. Although the Mössbauer method can be used for measurements of γ-line widths, the results are subject to considerable errors.

One of the most striking uses of the extreme energy resolution obtainable by the Mössbauer effect was achieved by R. V. Pound and G. A. Rebka, who measured the emission of photons in the direction towards the earth's center, and in the opposite direction from the earth's center. They found that the photon increased its energy by one part in 10^{16} per meter when falling in the earth's gravitational field. This can be taken as a proof that the photon of $E_{\mathbf{h}\nu} > 0$ does have a mass, cf. §2.3.

When a "Mössbauer pair" (like ^{191}Os/Ir, or ^{57}Co/Fe, ^{119m}Sn/Sn, ^{169}Er/Tm, etc.) have source and absorber in different chemical states, the nuclear energy levels differ for the two Mössbauer atoms by some amount ΔE_γ. By using the same technique as in Figure 7.25, resonance absorption can be brought about by moving the source with a velocity corresponding to ΔE_γ. In this manner, a characteristic *Mössbauer spectrum* of the compound (relative to a reference compound) is obtained; the location of the peaks (i.e. the absorption maxima) with respect to a non-moving source (the *isomer shift*) is usually given in mm s^{-1}.

Figure 7.27 shows the isomer shifts obtained for a number of actinide compounds. The positions of the isomer shifts show the effect of valence states due to different population of the 5f orbitals. The different shifts for compounds of the same valence state is a measure of the variation in the covalency of the bonding. The compounds on the left are metallic. The shifts reflects the contributions of conduction electrons to the electron density at the nucleus of neptunium.

Mössbauer spectroscopy is limited to the availability of suitable sources. About 70 Mössbauer pairs are now available. The technique provides a useful method for studying chemical compounds in the solid state, especially compounds which are nontransparent to light and chemically or radioactively unstable. Mössbauer spectroscopy has also been used by several Mars landers, e.g. Spirit and Opportunity, to analyse rocks and minerals on the surface of Mars looking for signs left from ancient surface water.

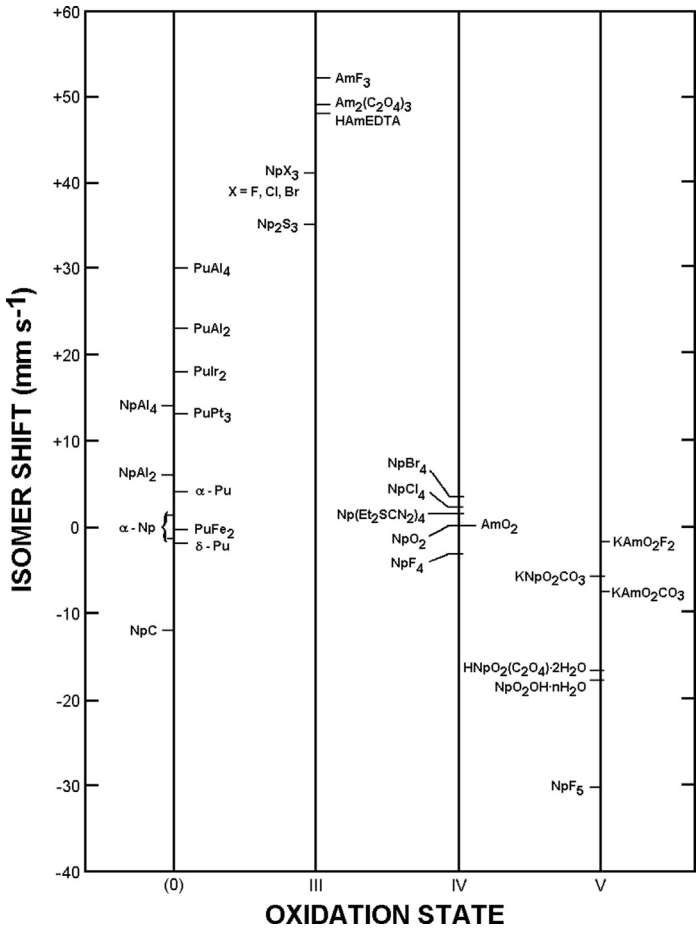


Figure 7.27 Isomer shifts for some actinide metals and actinide compounds.

7.9. TECHNICAL APPLICATIONS OF RADIATION SOURCES

Nuclear radiation absorption methods have many technical applications. These methods are not to be confused with radioisotope tracer methods, although radioisotopes may be used as radiation sources. In the tracer method the chemical properties of the radionuclide are important while in the applications discussed in this section only the type and energy of radiation emitted are important.

As a *source* of radiation in such technical applications, either accelerators or radiation from radionuclides can be used. Interchangeable radionuclides have the advantage over accelerators as radiation sources in that they can cover a larger energy range from high energy γ-rays to low energy β-rays in a much simpler way. This makes it possible to select the type and energy of radiation which have the most advantageous properties for a